docker

Lamba in a box

Real Integration tests with Localstack and Go

About me

You can find me as **@mdelapenya** everywhere

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Staff Software Engineer @ Docker
Computer Science degree, Master in Software Engineering

- Testcontainers Go maintainer since 2020
- Engineering Productivity at Elastic Observability
- QA Tech lead at Liferay Cloud
- Core Engineer at Liferay
- In OSS since 2011
- Hitting keyboards since 1994

What we are going to see today:

01. Meet the app
02. Your Local Dev Environment
03. Building and Deploying an AWS Lambda in Go
04. Testing with Localstack
05. Running Localstack with Testcontainers Go
06. Q&A

1. Meet the app

Meet the app

We have landed into a new project and need to get acquainted with it.

- ➔ Persistence layer, using Postgres as relational database.
- ➔ Cache layer, using Redis.
- ➔ Streaming events layer, using Redpanda.
- ➔ **Cloud layer, using AWS Lambdas to calculate some statistics.**

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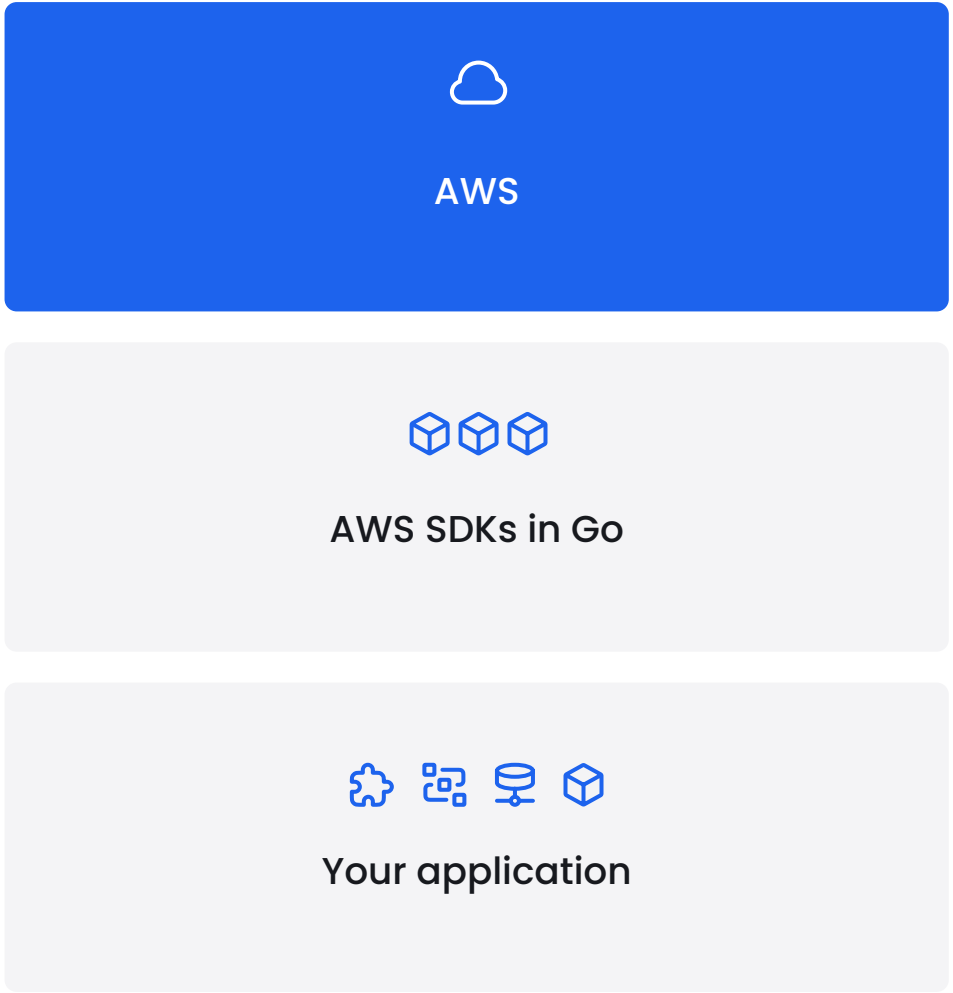
2. Your Local Dev Environment

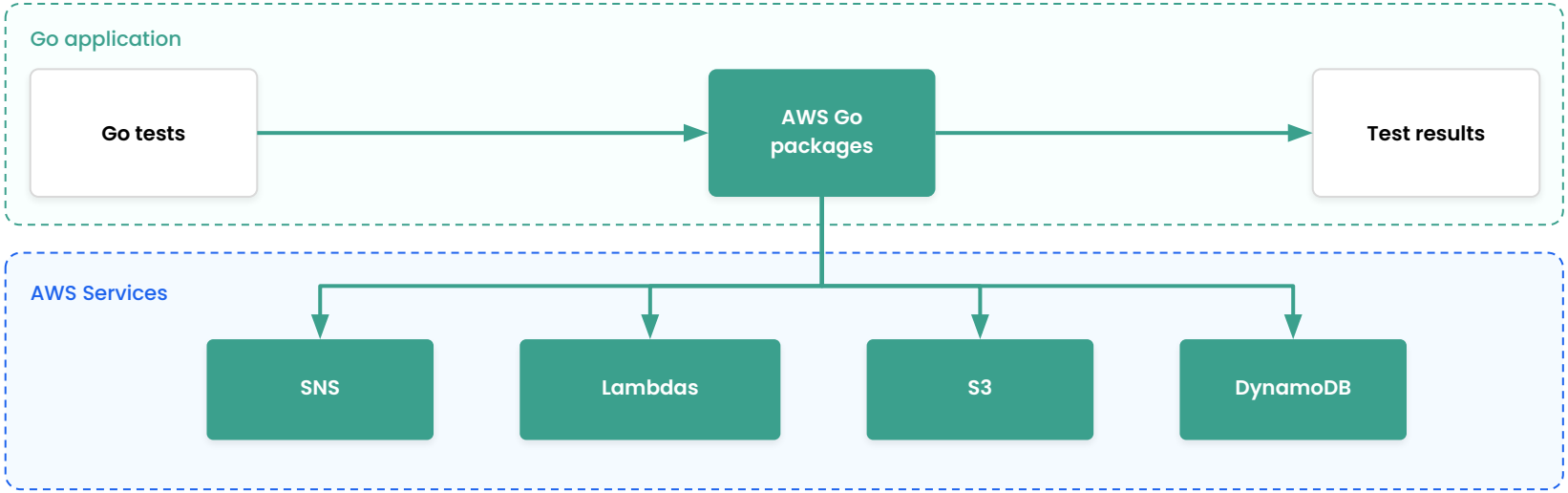
Develop with the Cloud

How it works

- Our company uses AWS as Cloud provider
- We setup AWS's SDKs into the project
- We configure the credentials
- We start coding...



Developing Cloud applications



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3. Building and Deploying and AWS Lambda in Go

Demo (1)

Build: https://docs.aws.amazon.com/lambda/latest/dg/golang-handler.html

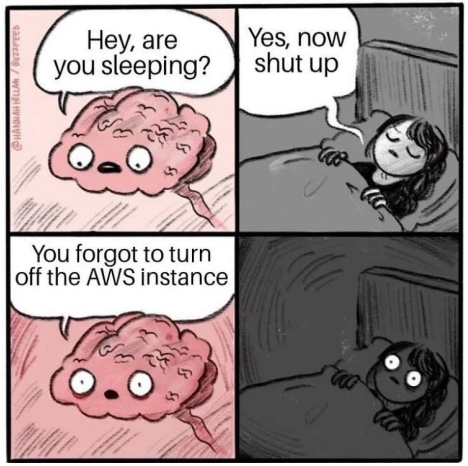
Deploy: https://docs.aws.amazon.com/lambda/latest/dg/golang-package.html

How do you test those cloud scenarios?

Testing Cloud applications

- ~~No tests, my code is perfect!~~
- Test environment in the Cloud provider
  - Per team?
  - Per developer?
  - Shared across the company?
  - How long does it take to have them?
  - Do you prune outdated resources?

Do you measure AWS costs?



Test Environments

AWS SDKs in Go

Your application

4. Testing with Localstack

Localstack

➔ https://localstack.cloud

➔ **Installation:** https://docs.localstack.cloud/getting-started/installation/

Loved by Developers

Backed by a large community and rooted in open source. Make cloud development more enjoyable by reducing some of the friction and wait times that sometimes come with it. Once tried, it's hard to go back to working without it.

300M+
Docker Pulls

2M+
Weekly Docker Pulls

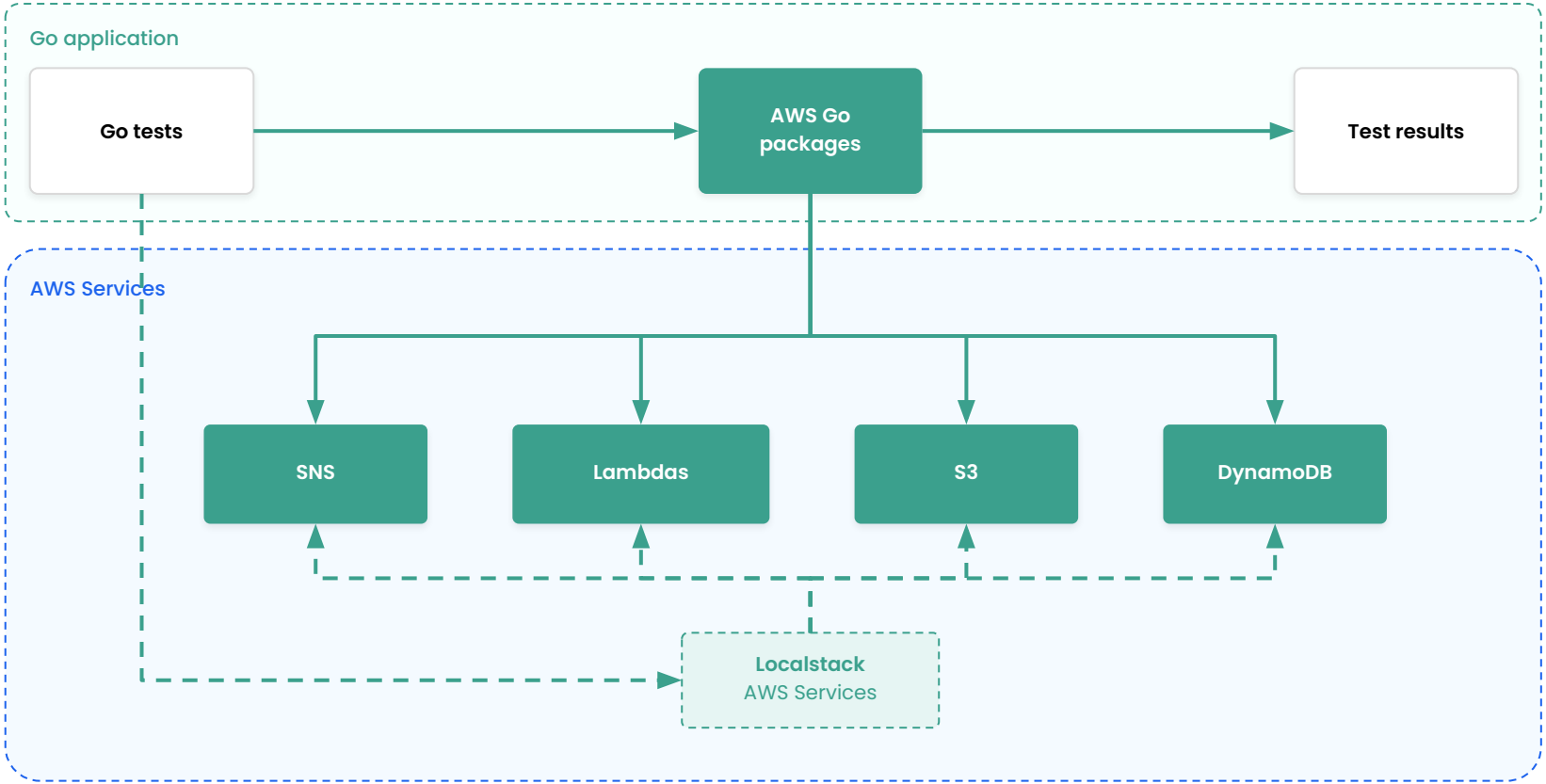
8M+
Weekly LocalStack Sessions

52k+
Github Stars

25k+
Slack Users

500+
Contributors

Testing Cloud applications with Localstack



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5. Running Localstack with Testcontainers for Go

Testcontainers Go

An Open Source Go package (MIT license) providing developer-friendly API's on top of the Docker engine.

https://github.com/testcontainers/testcontainers-go

- Start, stop, terminate containers and networks
- Wait for containers on custom conditions
- Lifecycle hooks to inject custom code (Pre/Post)
- Copy files to/from containers
- Garbage collection of Docker resources

Go package: docker/docker

Go package: testcontainers-go

Your Go app

Testcontainers Go: modules

Go packages providing access to the most used technologies:

- Cloud Emulators: **Localstack**, Google Cloud, Azure
- Relational DBs: Mysql, Postgres, ...
- Vector DBs: Weaviate, Chroma, Qdrant, Milvus, Pinecone...
- Non Relational DBs: Elasticsearch, Redis, MongoDB, Neo4j, Opensearch...
- Keycloak, OpenFGA, Vault...
- ~70 different Go modules!
- Convenient API specific to each module.

https://www.testcontainers.com/modules

Go package: testcontainers-go

Go packages: testcontainers-go/modules

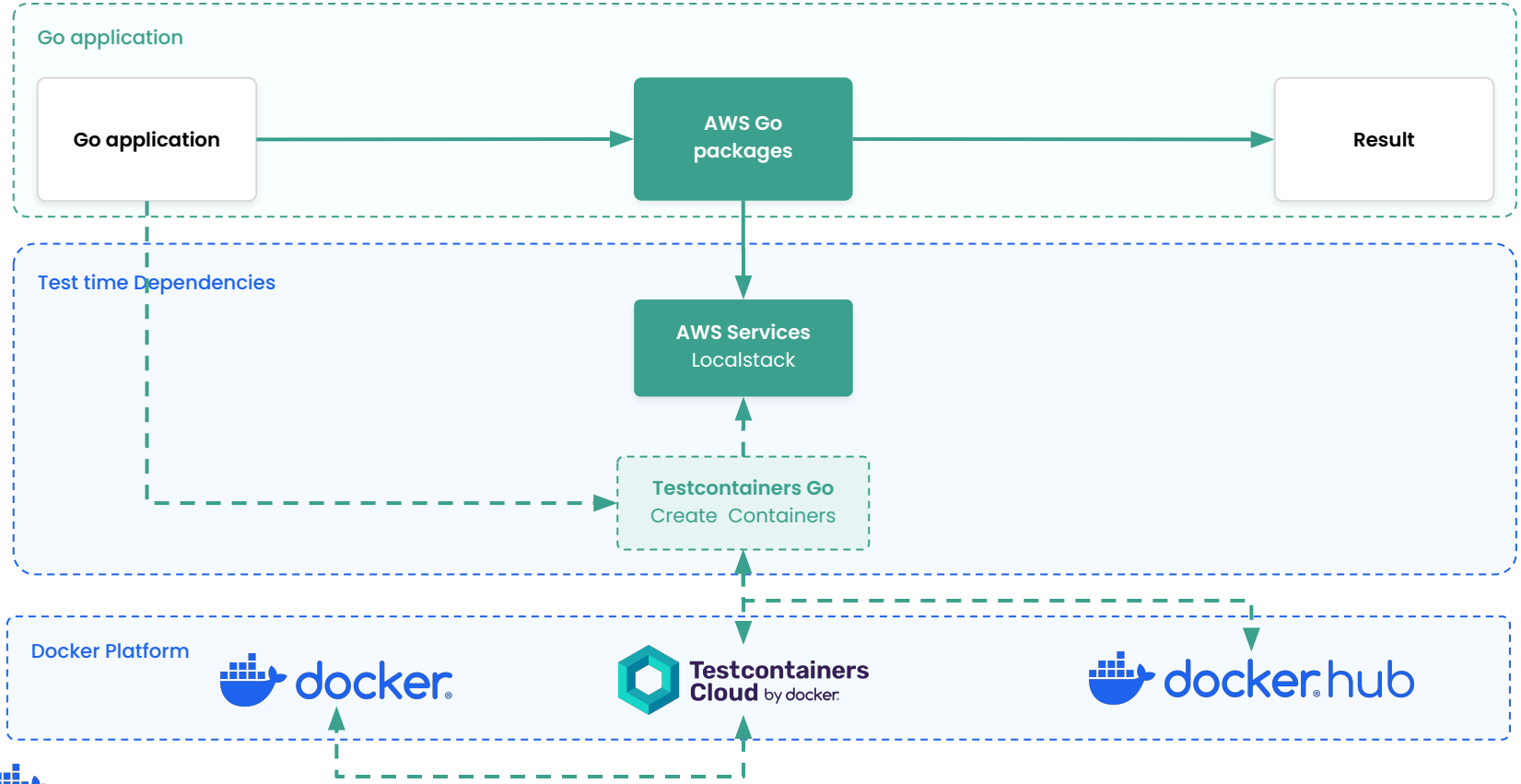
Your Go app

Testcontainers for Go & Localstack

A module exists! https://testcontainers.com/modules/localstack/?language=go

```go
localstackCtr, err := localstack.Run(ctx, "localstack/localstack:latest")
if err != nil {
	log.Printf("failed to start container: %s", err)
	return
}
```

Dockerised workflow



Demo (2)

Integration Tests:
https://github.com/testcontainers/workshop-go/blob/solutions/step-7-adding-localstack.md#adding-the-localstack-instance

Demo (3)

Local Dev Mode:
https://github.com/testcontainers/workshop-go/blob/solutions/step-4-dev-mode-with-testcontainers.md

6. Conclusions and Q&A

Conclusions

- **Testcontainers Go** + **Localstack**: a really powerful and easy-to-use combo for local development experience.
- **Integration tests** will give you enough confidence so you can make progress with speed, but you still need to test against the real thing, AWS.
  - Run lots of integration tests in your local machine
  - You still need to run **SOME E2E tests** against the real thing, within a less frequent schedule!
- **Dev Mode:** Testcontainers Go enables local development environments in code, just by leveraging init functions + build tags.

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Thank you!

Localstack Online Meetup

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Resources

https://github.com/testcontainers/testcontainers-go
https://github.com/testcontainers/workshop-go

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